Enhancing Visual Grounding in Remote Sensing with Scene-Level Guidance

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***Abstract*—Remote sensing visual grounding (RSVG) aims to localize targets in remote sensing images using natural language expressions, which is important for Earth observation (EO) applications that require the detection of arbitrary categories. However, existing methods rely on manually designed text prompts, which fail to capture the visual context inherent in remote sensing images, thereby limiting the detection performance. To address this issue, we propose a fine-tuning framework for RSVG with Scene-Aware Prompt Learning Module (SAPLM), which leverages scene-level visual features to generate text embeddings that align with the corresponding scene. This enables the model to focus on targets that are more likely to appear while reducing interference from irrelevant objects. Experimental results demonstrate that our method achieves state-of-the-art performance, with 92.5% AP50 on DIOR and 58.7% mAP on DOTAv2.0.**

***Index Terms*—cross-modality fusion, object detection, prompt learning, visual grounding for remote sensing (RSVG).**

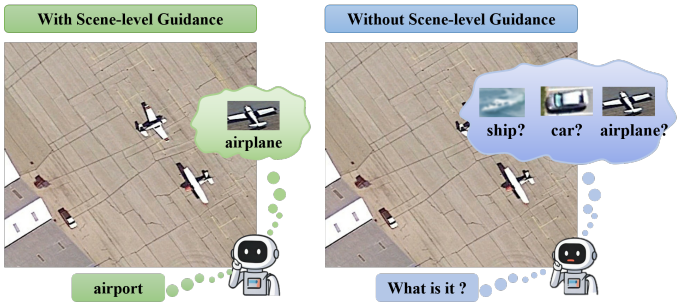

Fig. 1. **Motivation.** Incorporating visual scene-level guidance facilitates the model's understanding of remote sensing scenes. If we know that the current image depicts an airport, we expect that "ship" will not appear in the image, which makes it easier to detect "airplane".

I. Introduction

Remote sensing object detection is a fundamental task in Earth observation, playing a crucial role in applications such as urban planning, traffic monitoring, and disaster assessment. Because of the reliance on predefined category sets, traditional object detection methods often need to reconstruct a dataset and train a new model when the detection categories change, thus limiting their scalability in real-world remote sensing scenarios [1]. To address this issue, recent works [2], [3], [4] reformulate object detection as a visual grounding task, which localizes objects based on arbitrary natural language descriptions. This paradigm alleviates the constraint of fixed category sets and enables open-vocabulary detection.

Despite their success, existing visual grounding methods for remote sensing (RSVG) typically adapt models pre-trained on natural image datasets through fine-tuning on remote sensing data [5], [6]. However, they heavily rely on manually designed text prompts, which fail to capture the rich visual context

inherent in remote sensing images. As shown in Fig. 1, the concept of "airplane" is often associated with contextual elements such as "airport" or "runway". Providing a prompt containing the prior knowledge that the current image depicts an airport can assist the model in better analyzing the targets. Fixed prompts lack the ability to incorporate such prior knowledge, thereby limiting the model's understanding of the corresponding scene and affecting the detection performance.

Xu et al. [7] suggest that although natural language exhibits high information density and strong generalization capability, leveraging richer visual cues from the image can further enhance the text prompts. Similarly, LAE-DINO [8] introduces a visual-guided text (VisGT) module to incorporate image-level scene features into text embeddings. However, their approach directly aggregates visual features into text representations, neglecting the cross-modal alignment gap between vision and language, which may lead to suboptimal feature integration.

To overcome these limitations, we propose a novel fine-tuning framework for RSVG. It introduces Scene-Aware Prompt Learning Module (SAPLM), which dynamically refines text embeddings using scene-level visual features, producing scene-aware learnable text prompts tailored to the input image. Specifically, SAPLM extracts global scene representations from the visual backbone and injects them into text features to enhance cross-modal alignment. This enables the model to focus on objects that are more likely to appear in the given scene while suppressing irrelevant or misleading predictions, thereby improving detection performance.

II. Methodology

A. Brief Review of Grounding DINO

Visual grounding aims to localize object instances in an image based on natural language expression (e.g., "airplane. vehicle. ship. … windmill"), and the core challenge lies in establishing accurate alignment between visual and language features. Let $\mathbf{I} \in \mathbb{R}^{N \times D}$ denote the multi-scale visual features extracted from the image backbone [9], where N represents the number of visual patches and D represents the feature dimension. Similarly, let $\mathbf{T} \in \mathbb{R}^{S \times D}$ be the text embedding generated by the text backbone [10], where S is the length of the tokenized text sequence.

As shown in Fig. 2, Grounding DINO [3] first fuse $\mathbf{I}$ and $\mathbf{T}$ via its feature enhancer, which aligns visual and textual representations via cross-attention mechanisms. The fused features, denoted as $\widetilde{\mathbf{I}}$ and $\widetilde{\mathbf{T}}$, are then fed into the query selection module to generate cross-modal queries that are relevant to the input text description. Subsequently, the cross-modality queries interact iteratively with $\widetilde{\mathbf{I}}$ and $\widetilde{\mathbf{T}}$ in cross-modality decoder. Finally, a detection head predicts a set of bounding boxes and corresponding category labels based on the decoder output.

B. Scene-Aware Prompt Learning Module

Text prompts play an important role in vision-language models, but creating appropriate text prompts is extremely challenging due to the wide variety of these large scale aerial datasets, and the existence of domain-specific nuances. In order to distinguish the relevance of different text tokens to the image scenes, we propose Scene-Aware Prompt Learning Module (SAPLM) to adapt Grounding DINO to remote sensing detection. It enhances text prompt using scene-level visual feature before sending it to the feature enhancer.

As shown in Fig. 2, SAPLM extracts scene-level visual and text features, then aligns them through contrastive loss, these visual features will be integrated into the text features to generate learnable text prompts. First, we extract scene-level visual features based on Multi-scale Deformable Self-Attention (MSDA) [11]. In each layer, MSDA can adaptively select the image regions that need attention according to the characteristics of the input visual features, and then perform projection transformation through a fully connected layer as follows,

$$\mathbf{I}^{\mathrm{l}} = FFN^{l}(MSDA^{\mathrm{l}}(\mathbf{I}^{\mathrm{l-1}})),\ l = 1, 2, ..., L_i\ , \tag{1}$$

where $\mathbf{I}^0 = \mathbf{I}$ is the features extracted from image backbone, L_i is the number of layers for MSDA and FFN. Next, average pooling is applied to the output result of the last layer to extract visual scene features as,

$$\mathbf{I}_{\mathrm{sc}} = mean(\mathbf{I}^{L_i}, \mathrm{dim} = 0)\ . \tag{2}$$

Subsequently, the visual scene features $\mathbf{I}_{\mathrm{sc}}$ are fused into the text features $\mathbf{T}$. To further overcome the cross-modal semantic discrepancies between vision and text, a learnable linear projection is employed for intermodal alignment, which maps $\mathbf{I}_{\mathrm{sc}}$ to a semantic adjuster adapted to the current scene, and therefore imposing a controllable modulation on the text features $\mathbf{T}$. To further enhance the image-awareness of the adjusted text features, FFN is used to perform a nonlinear transformation,

$$\mathbf{T}^{\mathrm{l}} = FFN^{l}(\mathbf{T}^{\mathrm{l-1}} + \mathbf{I}_{\mathrm{sc}}\mathbf{W}_v^l),\ l = 1, 2, ..., L_{i2t}\ , \tag{3}$$

where $\mathbf{T}^0 = \mathbf{T}$ is features extracted from text backbone, L_{i2t} is the number of layers for adjustment, and $\mathbf{W}_v \in \mathbb{R}^{D \times D}$ is learnable linear projection matrix.

C. Loss Function

Following DETR-based works [11], [12], [13], we use L1 and GIoU loss [14] for bounding box loss, denoted as L_{loc}. For classification, contrastive loss L_{cls} between predicted objects and language tokens is used [2].

In addition, we employ contrastive loss to align the scene features of images and text. Given a training image, the corresponding text feature is $\mathbf{T} \in \mathbb{R}^{S \times D}$. Assume the text feature corresponding to the i -th class is $\mathbf{T}_{\mathbf{i}}^{'} \in \mathbb{R}^{S_i \times D}$. S_i denotes the token length of the text description of the i -th class, summing along this dimension results in the class-level text feature $\mathbf{T}_i \in \mathbb{R}^{1 \times D}$. According to eq (4), the weighted average of all positive text features is computed as scene-level text features $\mathbf{T}_{sc} \in \mathbb{R}^{1 \times D}$, which are used to align with the visual scene features $\mathbf{I}_{sc} \in \mathbb{R}^{1 \times D}$.

$$\mathbf{T}_{\mathrm{sc}} = \frac{1}{\mathrm{N_p}} \sum_i \mathrm{N_p^i} \mathbf{T}_{\mathrm{i}}\ , \tag{4}$$

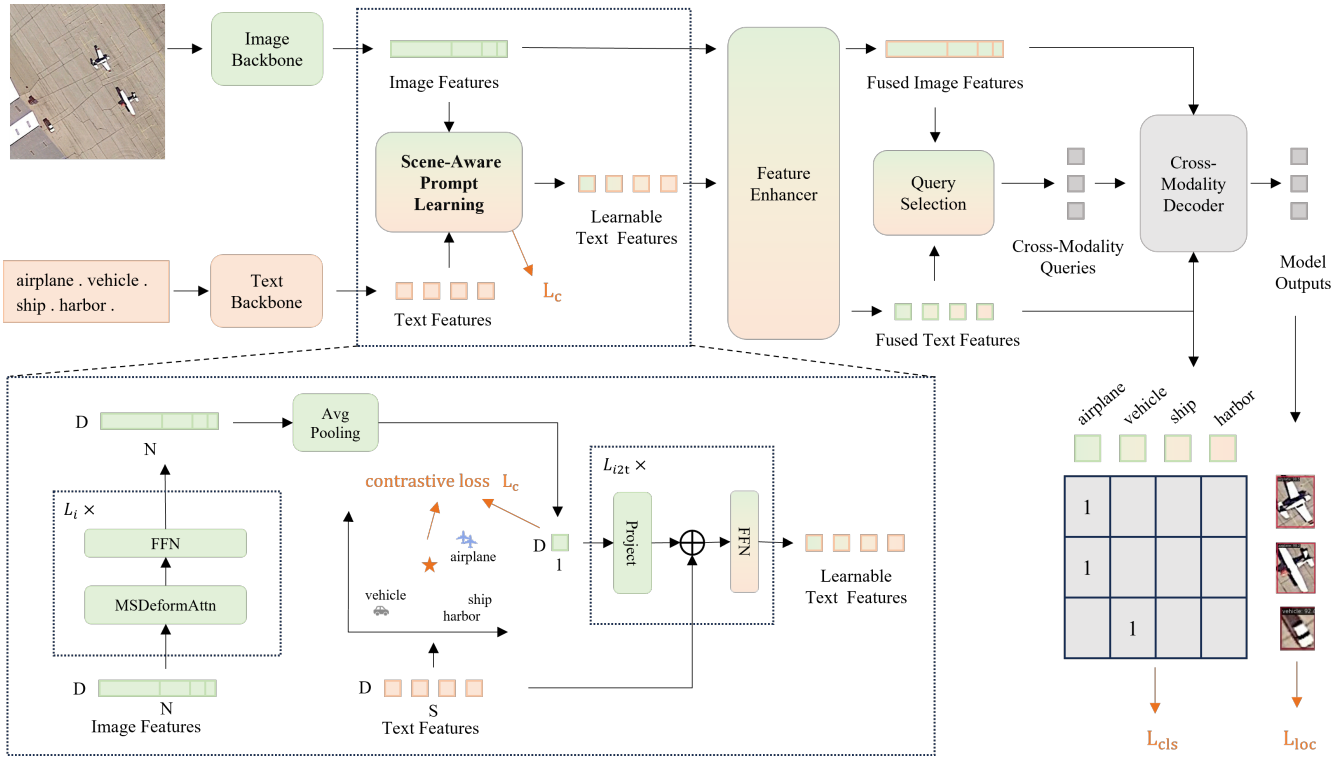

Fig. 2. **Overview of the proposed method.** The framework builds upon Grounding DINO and incorporates the Scene-Aware Prompt Learning Module (SAPLM), which leverages scene-level visual features to generate learnable text features for fine-tuning in specific datasets. The visual and text features are then processed by the feature enhancer and cross-modality decoder for final detection results.

where N_p denotes the total number of positive samples in the image, the weight N_p^i represents the number of positive samples for the i -th class, satisfying $\sum_i N_p^i = N_p$, $0 \leq N_p^i \leq N_p$. Given a batch of B positive pairs, their corresponding scene-level visual and text features are denoted as $\{ (\mathbf{I}_{sc}^1, \mathbf{T}_{sc}^1), (\mathbf{I}_{sc}^2, \mathbf{T}_{sc}^2), ..., (\mathbf{I}_{sc}^B, \mathbf{T}_{sc}^B)\}$. The cosine similarity is computed as $s_{i,j} = \mathbf{I}_{sc}^i \ \mathbf{T}_{sc}^{j\ T}$. Consequently, we obtain the vision-to-text contrastive loss and the text-to-vision contrastive loss, defined respectively as,

$$L_{vti} = -\log\left(\frac{\exp(s_{i,i}/\tau)}{\sum_{j=1}^{B}\exp(s_{i,j}/\tau)}\right) , \tag{5}$$

$$L_{tvi} = -\log\left(\frac{\exp(s_{i,i}/\tau)}{\sum_{j=1}^{B}\exp(s_{j,i}/\tau)}\right) , \tag{6}$$

where τ is the temperature parameter, and the contrastive loss can be calculated as

$$L_c = \frac{1}{2B}\sum_{i=1}^{B}(L_{vti} + L_{tvi}) . \tag{7}$$

The final loss function is

$$L = L_{cls} + \alpha L_{loc} + \beta L_c . \tag{8}$$

III. Experiments

A. Experimental Setup

1) Dataset: According to [8], we conducted experiments on the DIOR [15] and DOTAv2.0 [16] datasets to evaluate the effectiveness of the proposed method. DIOR consists of 23463 remote sensing images with 800×800 image resolution, including 20 object categories. The original size of images in DOTAv2.0 ranges from 800×800 to 4000×4000, including 18 categories. Although orient bounding boxes are provided in DOTAv2.0 dataset, we only use horizontal box annotations for training and testing.

2) Implementation Details: The experiments were conducted on a single NVIDIA RTX 4090 GPU. AdamW optimizer [17] is used with an initial learning rate of 1×10^{-4} , a weight decay of 1×10^{-4} , and the batch size is set to 4 for training.

In SAPLM, the number of layers L_i is set to 6 for MSDA and FFN, the number of layers L_{i2t} is set to 3 for fusing scene-level image features and text features, and the hyper-parameters α and β of the loss function are set to 1 and 10, respectively.

B. Comparison Results

Table I presents the comparison results with other visual grounding models. Specifically, YOLO-World [4], GLIP [2], GroundingDINO [3], LAE-DINO [8], and RT-OVAD [18] are open-vocabulary object detection models pre-trained on the large-scale LAE-1M [8] dataset, while GroundingDINO-FT and LAE-DINO-FT are models fine-tuned on specific datasets. It shows that our method achieves state-of-the-art performance on both DIOR and DOTAv2.0 benchmarks, surpassing the

TABLE I
COMPARISON WITH STATE-OF-THE-ART VISUAL GROUNDING METHODS ON THE DIOR AND DOTAV2.0 BENCHMARK TESTS. THE BEST RESULTS IN EACH GROUP ARE DISPLAYED IN BOLD.

Method	Year	DIOR (AP50)	DOTAv2.0 (mAP)
Models Pre-Trained on Remote Sensing Dataset LAE-1M			
YOLO-World	2024	78.0	33.5
GLIP	2022	82.8	43.0
GroundingDINO	2024	83.6	46.0
LAE-DINO	2025	85.5	46.8
RT-OVAD	2024	87.7	53.8
Models Fine-Tuned on Specific Datasets			
GroundingDINO-FT	2024	90.4	54.0
LAE-DINO-FT	2025	92.0	55.5
Ours	–	**92.5**	**58.7**

TABLE II
ABLATION STUDY ON DIFFERENT METHODS OF INTEGRATING SCENE-LEVEL VISUAL FEATURES INTO TEXT PROMPTS.

Method	DIOR (AP50)
Baseline (GroundingDINO-FT)	90.4
add	92
gated-fuse	92.2
SAPLM	**92.5**

previous best approach by 0.5% and 3.2% in terms of AP50 on the DIOR dataset, and mAP on DOTAv2.0, respectively. This improvement demonstrates the effectiveness of the proposed method, which enhances the understanding of visual scene context by SAPLM, and thus improves the detection performance.

C. Ablation Study

o evaluate different strategies for fusing visual scene features $\mathbf{I}_{sc}$ with text features $\mathbf{T}$, we conduct an ablation study and show the results in Table II. The simple addition strategy taken by [8] achieves an improvement of 1.6% over the baseline, which proves the effectiveness of incorporating scene information on the detection performance. Using gated fusion module [19] to fuse $\mathbf{I}_{sc}$ and $\mathbf{T}$ achieves an accuracy of 92.2%. Our proposed SAPLM module, which employs a simple projector and FFN, achieves the highest performance at 92.5% AP50, confirming that cross-modal alignment is necessary in adjusting text features.

IV. CONCLUSION

Visual grounding has enabled open-vocabulary object detection, but previous models overlook the importance of visual contextual information. In this paper, we introduce a fine-tuning framework for remote sensing images, which employ Scene-Aware Prompt Learning Module (SAPLM). It refines text embeddings using scene-level visual features to produce embeddings better suited for the specific remote sensing scenes. Experiments on DIOR and DOTAv2.0 show that SAPLM improves cross-modal alignment between images and text, achieving state-of-the-art detection performance on both datasets.

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